

Figure 7: Synthetic 3D Combination Data Example.

3D Combination The 3D combination task, evaluates the ability to mentally deconstruct a complex 3D object into its constituent parts and then identify which of those parts could be used to build a different target shape.

The task generation proceeds as follows: **Shape Generation:** A complex 3D shape is created, which serves as the "source" object. This source object is rendered from two opposite isometric viewpoints to give the user a complete understanding of its structure. **Component Segmentation:** The source object is programmatically broken down into a set of smaller, non-overlapping 3D components. These components are the basic building blocks that could theoretically form the original shape. **Question Formulation:** The "question" is presented as a new, different "target" 3D shape. **Option Generation:** The options provided to the user are the individual 3D components that were segmented from the original source object. These components are laid out individually for clear inspection.

Algorithm 4: GENERATE3DFOLDINGQUESTION

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1: Input:  $q\_id$ , difficulty tier list  $\{easy, medium, hard\}$ , number of distractors  $n$ 
2:  $seed \leftarrow \text{Hash}(q\_id)$ 
3:  $\text{SetRandomSeed}(seed)$ ;  $\text{ClearScene}()$ 
4:  $cube, faceMap \leftarrow \text{CreateCubeWithTextures}(seed=seed)$ 
5:  $layout \leftarrow \text{RandomChoice}(\{cross, T\})$ 
6:  $net \leftarrow \text{UnfoldCube}(cube, layout)$ 
7:  $\text{SetCamera}(\text{Top})$ ;  $\text{RenderImage}(q\_id\_Q)$ 
8:  $bestView \leftarrow \text{Best3DView}(cube)$ 
9:  $\text{SetCamera}(bestView)$ 
10:  $\text{RenderImage}(q\_id\_A0)$ 
11: for  $i \leftarrow 1$  to  $n$  do
12:    $tier \leftarrow \text{SelectTier}(i, n)$ 
13:    $dCube \leftarrow \text{CreateCubeDistractor}(cube, tier=tier, seed+i)$ 
14:    $\text{SetCamera}(bestView)$ 
15:    $\text{RenderImage}(q\_id\_A\{i\})$ 
16: end for
17:  $\text{SaveMetadata}(\{q\_id, seed, layout, faceMap\})$ 

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3D Folding The 3D box folding task evaluates a person's spatial reasoning ability, specifically their capacity to visualize how a 2D pattern (a "net") will fold into a 3D cube. The process for generating a question is as follows:

Cube and Texture Generation: A standard 3D cube is created. Each of its six faces is assigned a unique texture or color. This is the "target" cube.

Unfolding: The textured 3D cube is computationally "unfolded" into a 2D net. The net is a flat pattern that shows all six faces of the cube connected in a way that it could be folded back up into the cube. Common net patterns like a "cross" or "T-shape" are used. This 2D net serves as the "question" image.

Option Generation: A set of 3D cubes is then presented as the answer options.

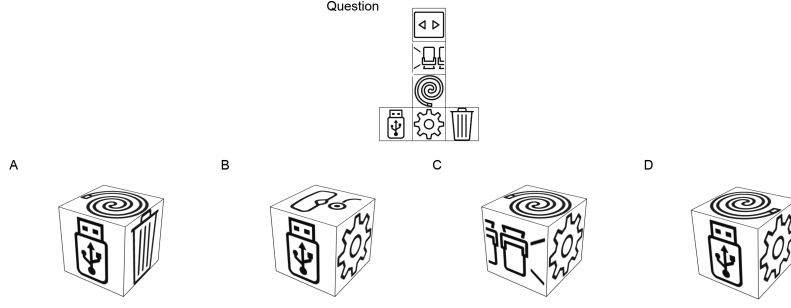


Figure 8: Synthetic 3D Folding Data Example.

The Correct Answer: This is a 3D rendering of the original, correctly folded cube, showing how the face textures are oriented in relation to each other.

Distractors: These are 3D cubes that are almost correct but have one or more faces manipulated in a way that makes the folded result incorrect. These manipulations can include:

Face Rotation: One or more faces on the cube are rotated from their correct orientation. Face Swapping: The positions of two or more faces are swapped. Texture/Color Replacement: The texture or color of one face is replaced with that of another.

Algorithm 5: GENERATESHAPEFINDINGQUESTION

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1: Input: question id  $q\_id$ , difficulty  $\in \{\text{easy}, \text{medium}, \text{hard}\}$ , options  $m = 4$ 
2:  $seed \leftarrow \text{Hash}(q\_id)$ 
3:  $\text{SetRandomSeed}(seed)$ ;  $\text{ClearScene}()$ 
4:  $cube \leftarrow \text{CreateCubeWithTextures}(seed)$ 
5:  $(V_0, V_1, V_2) \leftarrow \text{ChooseDistinctViews}(cube, 3, 120^\circ)$ 
6: for  $j \leftarrow 0$  to 1 do
7:    $\text{SetCamera}(V_j)$ ,  $\text{RenderImage}(q\_id\_V\{i\})$ 
8: end for
9:  $vis \leftarrow \text{VisibleFaces}(cube, V_2)$ 
10:  $f^* \leftarrow \text{SampleFace}(vis, \text{strategy}=\text{difficulty})$ 
11:  $mat_{orig} \leftarrow \text{GetMaterial}(cube, f^*)$ 
12:  $\text{SetMaterial}(cube, f^*, \text{Blue})$ ,  $\text{SetCamera}(V_2)$ ,  $\text{RenderImage}(q\_id\_V2)$ 
13:  $\text{SetMaterial}(cube, f^*, mat_{orig})$ 
14:  $opts \leftarrow \{f^*\} \cup \text{Sample}(\text{OtherFaces}(cube, f^*), m - 1)$ 
15:  $opts \leftarrow \text{Shuffle}(opts)$ 
16: for  $k, f$  in  $\text{Enumerate}(opts)$  do
17:    $\text{SetCamera}(\text{FaceNormalView}(cube, f))$ 
18:    $\text{RenderImage}(q\_id\_O\{i\})$ 
19: end for
20:  $\text{SaveMetadata}\{\text{id}:q\_id, seed, \text{views}:[V_0, V_1, V_2], \text{replaced}:f^*, \text{correctIdx}:\text{IndexOf}(f^*, opts)\}$ 

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3D Shape Finding The 3D Shape Finding task is a visual memory and attention task that tests the ability to track a specific face of a 3D object as the object is rotated in space. Here is how a typical question is generated: **Cube Generation:** A 3D cube is created with a unique, distinct texture applied to each of its six faces. **View Sequence:** The participant is shown a sequence of images (typically two) of the cube from different viewpoints. This allows them to see the cube and the arrangement of its face textures from multiple angles. The "Change" Event: A third image of the cube is then presented. In this view, one of the visible faces of the cube has its texture replaced with a solid color (e.g., blue). This is the key event in the task. **The Question:** The participant is implicitly asked: "Which of the original face textures was replaced by the solid color?" **Option Generation:** The answer options are a set of images, each showing one of the original, individual face textures from the cube. **The Correct Answer:** This is the image of the face texture that was replaced by the solid color in the third view. **Distractors:** These are the other original face textures from the cube.

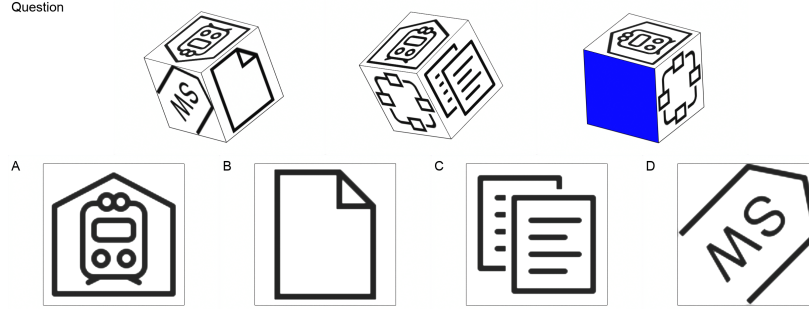


Figure 9: Synthetic 3D Shape Folding Data Example.

Task Name	Distractor Generation Logic	Specific Implementation Details
3D Rotation	A new 3D shape is created that is structurally different from the original, yet visually similar, testing the ability to spot subtle structural changes despite viewing-angle differences.	A new shape is generated by altering the “growth history” of the original object (e.g. adding or removing a block in a different location), producing a plausible but incorrect alternative.
3D Projection	An incorrect 2D orthographic projection is generated, testing the ability to accurately project a 3D object onto a 2D plane.	Generating a projection from an incorrect viewpoint (e.g. providing a <i>side view</i> when the <i>top view</i> was requested). Generating a projection of a slightly modified (distractor) 3D shape.
3D Combination	A valid component from the original shape is structurally modified, testing detailed analysis of part geometry.	A single authentic component is duplicated and then altered—typically by adding or removing a block—yielding a visually similar part that would not fit correctly into the complete assembly.
3D Folding	The 2D net is “folded” into an incorrect 3D cube, testing the ability to track face orientation and adjacency during folding.	Rotation: A face’s texture is rotated by 90°, 180°, or 270°. Swapping: Textures between two faces are swapped. Flipping: A texture is flipped horizontally or vertically.
3D Shape Finding	The options presented are the other, non-target faces of the cube, testing visual working memory and attention.	The task is to identify the original texture of a face that was replaced by a solid colour; distractors are the original textures of the other cube faces that were <i>not</i> the replacement target.

Table 8: Summary of Distractor Generation Logic